Dana Ferranti

Contact Information

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Current position

Worcester Polytechnic Institute Department of Mathematical Sciences
Assistant Research Professor — 2023 -
Advisor: Sarah Olson

Research interests

- Computational methods for fluid-structure interaction at low Reynolds number, particularly the method of regularized Stokeslets.
- Biological applications of Stokes flows.
- Reduced ODE models for systems of hydrodynamically coupled bodies.

Education

Tulane University, New Orleans, LA — 2017–2023

PhD, Mathematics
Thesis: *Regularized Stokeslet surfaces and a coupled oscillator system in Stokes flow*
Advisor: Ricardo Cortez

Clark University, Worcester, MA. — 2010–2014

BA, Mathematics and Computer Science, Magna Cum Laude

Awards

Outstanding Thesis Award (Monetary award) — 2024
Tulane University Department of Mathematics

Outstanding Graduate Instructor Award (Monetary award) — 2023
Tulane University Department of Mathematics

Publications

Pending

- Analysis of the stability of an immersed elastic surface using the method of regularized Stokeslets. (2025). (**DF**, S.D. Olson). Preprint: https://arxiv.org/abs/2507.07063. *Accepted pending minor revisions by Journal of Computational Physics.*

Published

- Regularized Stokeslet surfaces. *Journal of Computational Physics.* Volume 508, 113004 (2024). (**DF**, R. Cortez).
- Generalized matching preclusion in bipartite graphs. *Theory and Applications of Graphs.* Volume 5, Iss. 1, Article 1 (2018). (Z. Wheeler, E. Cheng, **DF**, L. Liptak, K. Nataraj).
- The value of prior knowledge in machine learning of complex network systems. *Bioinformatics.* Volume 30, 22 (2017). (**DF**, D. Krane, D. Craft).

Research Experience

- **Worcester Polytechnic Institute** — 2023 -
  Department of Mathematical Sciences
  - Stability analysis of a coupled elastic surface-fluid system using the method of regularized Stokeslets.
  - Generalized image systems for Stokes flows using the Lorentz reflection principle.
  - Mathematical modeling of cellular functions.
- **Tulane University** — 2017-2023
  Center for Computational Science in Mathematics Department
  - Extending the method of regularized stokeslets by using exact integration over triangulated surfaces.

- Reduced models of cilia interaction to investigating the potential effect of elastic coupling and inertia on synchronization.

- **Massachusetts General Hospital** 2016–2017
  Physics Research in Department of Radiation Oncology
  - Using theoretical models to demonstrate the value of prior knowledge in determining causal relationships in complex networks, with applications to machine learning in medicine.
  - Advisor: David Craft.

Undergrad. Research Advising

At Worcester Polytechnic Institute

- (2025 -) Second year aerospace engineering student. Topic: dynamical systems arising from coupled oscillators in unsteady Stokes flow.
- (2025) Third year physics student. Topic: Incorporating bending/elastic stiffness terms in the regularized Stokeslet surfaces framework.

Teaching Experience

As instructor
Worcester Polytechnic Institute

- Calculus IV — Fall 2025
- Independent Study: Introduction to the Immersed Boundary Method — Spring 2025
- Calculus II — Spring 2025
- Calculus IV — Spring 2024

Tulane University

- Probability & Statistics I (Math 1110) — Spring 2023
- Introduction to Applied Math (Math 2240). — Fall 2021

As teaching assistant
Tulane University

- Introduction to Applied Math (Math 2240). — 2019, 2020, 2021
- Linear algebra (Math 3090). — 2020
- Calculus I (Math 1210). — 2017, 2019
- Calculus II (Math 1220). — 2018, 2020
- Calculus III (Math 2210). — 2018

Service and Outreach

- Reviewer for *Pi Mu Epsilon* journal — 2024
- Volunteer at WPI's Sonia Kovalevsky Day — 2024
- President of AMS Graduate Student Chapter — 2019-2021
- Mathematics department tea time organizer — 2018-2022
- Treasurer of AMS Graduate Student Chapter — 2017-2019
- Member of Inclusivity in Mathematics Task Force at Tulane (IMTF) — 2020-2023

Posters & Talks

Posters

- *Stability Analysis of an Immersed Elastic Surface Using the Method of Regularized Stokeslets*
  - Mathematical Modeling, Computational Methods, and Biological Fluid Dynamics at National Institute for Theory and Mathematics in Biology (NITMB) in Chicago, IL (July 23, 2025).
  - Frontiers in Applied & Computational Mathematics at NJIT (June 5, 2025).
- *Inertial Effects in a Minimal Model of Hydrodynamically Dynamically Coupled Cilia* Blackwell-Tapia Conference at Institute of Mathematical and Statistical Innovation (IMSI) in Chicago, IL (Nov. 19, 2021).

Talks

- *Numerical stability analysis of a coupled fluid-elastic structure system using the method of regularized Stokeslets*
  78th Annual Meeting of the Division of Fluid Dynamics in Houston, TX (November 24, 2025)
- *Low Reynolds Number Fluid Dynamics: Physics Meets Biology Meets Computational Science*
  Xavier University of Louisiana, Math @ Xavier Talk (November 17, 2025)
- *Numerical stability analysis of a coupled fluid-elastic structure system using the method of regularized Stokeslets*
  Tulane University Applied and Computational Mathematics Seminar (November 14, 2025)
- *A Tutorial on the Method of Regularized Stokeslets: Advanced Methods*
  Mathematical Modeling, Computational Methods, and Biological Fluid Dynamics at NITMB in Chicago, IL (July 21, 2025).
- *Numerical stability analysis of a coupled fluid-elastic structure system using the method of regularized Stokeslets*
  77th Annual Meeting of the Division of Fluid Dynamics in Salt Lake City, UT (November 25, 2024).
- *Linear stability analysis of a coupled fluid-structure system using the method of regularized Stokeslets*
  Joint annual meeting of Korean Society for Mathematical Biology and Society for Mathematical Biology (July 1, 2024).
- *Regularized Stokeslet Surfaces*
  77th Annual Meeting of the Division of Fluid Dynamics in Washington D.C. (November 20, 2023).
- *Simulating bodies immersed in viscous flows: new developments in the Method of Regularized Stokeslets (MRS)*
  Worcester Polytechnic Institute Mathematics Colloquium (September 8, 2023).
- *Regularized Stokeslet Surfaces* Scientific Computing Around Louisiana (March 11, 2023).
- *Regularized Stokeslet Surfaces*
  Math for All in NOLA (February 25,2023).
- *An Extension to the Method of Regularized Stokeslets*
  Special session on Recent Developments in Numerical Methods for PDEs, Joint Math Meetings 2023 (January 4,2023).
- *Computational Modeling of Bodies Immersed in Viscous Fluids*
  Hunter College Applied Math Seminar (November 3,2022).